

Motivation letter (draft)

Draft status: passages marked [TODO: ...] require your real biography — fill them in before sending. Nothing in brackets should survive to the sent version. Target length: one page.

Dear members of the selection committee,

I am writing to apply for the PhD position at the Music Technology Group in collaboration with Steinberg Media Technologies, on representations and prediction of DAW states in music production.

[TODO: one short paragraph — who you are: degree(s), current role, and the one line of research or engineering experience most relevant to ML + audio. Be concrete: institution, year, topic.]

My motivation for this topic is practical before it is academic. [TODO: 2–3 sentences on your own music production background — DAWs you work in, what you produce, and a concrete moment where you felt the gap this PhD addresses: assistance tools that don't see the session, only the audio.]

That experience shaped the conviction at the core of this application: the DAW session itself — tracks, chains, routing, edits — is where production knowledge lives, and it deserves to be a first-class object of study. To test whether this conviction survives contact with real engineering, I built **Session State Explorer** ^{vo}, a working prototype that represents Ableton-style session state as a typed graph, links it to audio descriptors, and generates recommendations that are explainable by construction — each suggestion cites the graph nodes it reasons over and states its own limits. It also includes a first learned baseline for the call's central goal, predicting device-chain composition from session context. The prototype, its documentation, and a preliminary thesis proposal accompany this application.

I am applying to the MTG specifically because the group's identity — rigorous MIR grounded in open science and a stated commitment to trustworthy, transparent AI for music — matches how I believe assistive production research must be done. Essentia and Freesound are the natural substrate for the descriptor and material layers of this work, and the chance to develop the Cubase/VST3 instantiation of session-state graphs with Steinberg makes the industrial half of the project concrete in a way few PhD settings could.

[TODO: one paragraph — evidence you can execute: ML/DSP coursework or projects, Python engineering depth, any publications, teaching, or collaborative/interdisciplinary work. Tie each item to the candidate-profile bullets in the call.]

I would welcome the opportunity to discuss the proposal and the prototype. Thank you for your consideration.

Sincerely, Zachary Scheffler zachscheffler@gmail.com ·
github.com/colonelkernel/AbletonSessionStateExplorer